Research on the Application of Digitalization Methods in the Construction of Media Technology Laboratories

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Abstract

In the rapidly developing digital age, media technology laboratories actively embrace the development of digitization and take effective measures to promote the reform of digital education in media technology. Media technology laboratories bear the task of cultivating professional talents in digital media, and the demand for their digital construction is becoming increasingly urgent. This paper, combined with the development background of digital construction of media technology laboratories, analyzes the significance of its digital construction and proposes ideas and specific implementation methods for the digital construction of media technology laboratories. The implementation method includes the construction of a digital system for laboratory resources and a digital system for teaching methods, ultimately achieving comprehensive digitalization of media technology education.

CCS Concepts

• **Information systems** → Data management systems; Information integration.

Keywords

Digitalization, Media Technology Laboratories, Construction

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1 Introduction

Now digitalization has become the mainstream trend of today's society. As an important base for cultivating innovative talents and promoting the transformation of scientific research achievements, the digital construction of media technology laboratories is particularly important[1].The digital construction of media technology laboratories can not only significantly improve the laboratory's operational efficiency and management level but also enhance the quality of teaching, promote scientific and technological innovation, and thus have a profound impact on talent cultivation. Through digital means, media technology laboratories can break through the spatial and temporal limitations of traditional teaching, expand learning space and knowledge reserves, stimulate enthusiasm for learning and research, and make a greater contribution to the cultivation of innovative talents and the promotion of scientific and technological progress.

In recent years, with the rapid development of digital technology, it has had a profound impact on various fields such as digital technology, media, and art design. The discipline of digital media technology is an interdisciplinary field that integrates information technology, cultural creativity, and artistic creativity[2], gradually occupying an important position in fields such as media, entertainment, and education. To adapt to this trend, media technology laboratories need to carry out digital construction to better study and apply digital media technology and promote the development of related industries.

With the continuous innovation and reform of educational models, traditional teaching methods and means have become difficult to meet the needs of modern education. Universities, as the cradle of talent cultivation, need to actively embrace digitalization and promote the digital transformation of education and teaching. As an important platform for practical teaching in universities, the digital construction of media technology laboratories helps to improve the quality of teaching and cultivate students' innovative and practical abilities.

Digital construction helps to achieve the sharing and opening up of laboratory resources[3]. By building a digital platform[4], sharing of experimental equipment, experimental data information, and improving the utilization efficiency of laboratory resources can be realized. The digital platform can provide more convenient and efficient services, including online reservation and remote operation, thereby attracting more researchers to use laboratory resources and further promoting the development of resource sharing and opening up.

By introducing advanced information management systems, laboratories can achieve comprehensive digital management of equipment, data, information, etc., thereby optimizing resource allocation, reducing manual operations, lowering management costs, and improving the overall operational efficiency of the laboratory. At the same time, digital construction also helps to standardize and standardize laboratory business processes, improving the accuracy and reliability of laboratory work. Building a digital service platform for laboratories can change in real time with the needs of

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scientific and technological innovation, aggregating relevant digital resources to rapidly develop and take a leading edge in new fields. Digital construction is conducive to cultivating first-class dynamic innovation capabilities, effectively enhancing the organization's collaborative innovation capabilities in cutting-edge fields that require interdisciplinary knowledge at a high level, thereby improving the overall core competitiveness of the research team.

2 Ideas of the System

Compared to the previous digital construction of media technology laboratories that often only had a single platform or a few unrelated platforms, the digital construction method proposed in this paper is composed of two comprehensive systems, and the platforms within each system are interrelated and mutually supportive. This enables the digital construction of the media technology laboratory to form a closed loop as a whole, with the system possessing strong and complete functions, and also provides a new perspective for future research.

2.1 Laboratory Resource Digitalization Software System

Through a comprehensive review of the various resources of the media technology laboratory, including computer rooms, digital imaging, virtual simulation equipment, etc., resources are classified by type and function to achieve resource integration and optimized allocation. A digital resource platform for the laboratory is established to store laboratory resources in a digital form, ensuring data accessibility and security. Regular data backups of the resource library are conducted to prevent data loss or damage. Developing a digital software system is an important part of the digital construction of the media technology laboratory[5].The Laboratory Resource Digitalization Software System is shown in Figure 1 and mainly includes the following platforms:

(1).Digital Management Platform for Laboratory Equipment: This platform provides functions such as equipment inventory, student reservation applications, and equipment repair requests, meeting the daily needs of experimental teaching and ensuring the integrity of laboratory equipment.

(2).Digital Management Platform for Laboratory Classrooms: Laboratory administrators can log in to the platform to maintain information about the laboratory classrooms, and students can log in to the platform to make reservations for laboratory classrooms.

(3).Digital Monitoring Platform for Laboratory Environment: A fully digital monitoring platform is constructed by networking smart cameras and deploying intelligent sensors. The software system judges the monitored thresholds, and if conditions are triggered, it sends alarm notifications to the administrators via SMS, phone calls, or WeChat. At the same time, laboratory management personnel can log in to the digital monitoring platform to understand the environmental indicators and the use of various resources in the laboratory in real time, facilitating the timely discovery of problems.

(4).Laboratory System Management Platform: The platform provides functions such as adding or deleting laboratory users, configuring user menu permissions, and allows different user roles to operate different platform modules. It also offers the functionality for basic data reporting and export, making it convenient for administrators to collect data from various platforms and analyze and utilize the data promptly.

2.2 Digital Teaching Method System

The digitization of teaching methods has become increasingly feasible with the in-depth application of digital and intelligent technologies in the field of education. A digital system for teaching methods analyzes students' learning data to provide targeted teaching suggestions for teachers and personalized learning paths for students. The process of the digital system's operation is depicted in Figure 2 as follows:

(1)The digital system collects students' learning data, including multi-dimensional data such as homework completion status, classroom participation, and online learning duration.

(2)Based on big data analysis and artificial intelligence technologies, the collected data is processed to remove confounding information and extract data related to learning behavior using AI algorithms[6], effectively enhancing the accuracy of the data.

(3)The digital system recommends personalized learning paths for students based on the processed results. Personalized learning path recommendations are based on the analysis of students' learning data, catering to the unique learning styles and paces of different individuals by recommending suitable learning resources and courses. This not only stimulates students' interest in learning but also helps them grasp knowledge more quickly and improve learning efficiency.

(4)The digital system provides teachers with targeted teaching suggestions based on the results of intelligent technology processing. These suggestions cover aspects such as teaching methods, content, and pace, providing a basis for teachers to develop new teaching strategies.

(5)Based on the data analyzed and provided by the digital system, teachers can more accurately identify students' weaknesses and strengths, and thus devise teaching plans that are more aligned with the actual conditions of the students.

3 System Implementation

3.1 Implementation of the Laboratory Resource Digitalization Software System

3.1.1 System Framework. As previously mentioned, the laboratory resource digitalization software system mainly includes the following modules: the digital management module for laboratory equipment, the digital management module for laboratory classrooms, the digital monitoring module for the laboratory environment, and the laboratory system management module. The entire software system adopts a three-tier technical framework as shown in Figure 3, which is divided into the data layer, the service layer, and the application layer. The data layer primarily utilizes databases such as MySQL, MongoDB to store entity data for experiments, and Redis to store data that is frequently accessed and not often changed. The entity data within this system covers all the necessary data required for the four sub-modules, as detailed in Table 1.

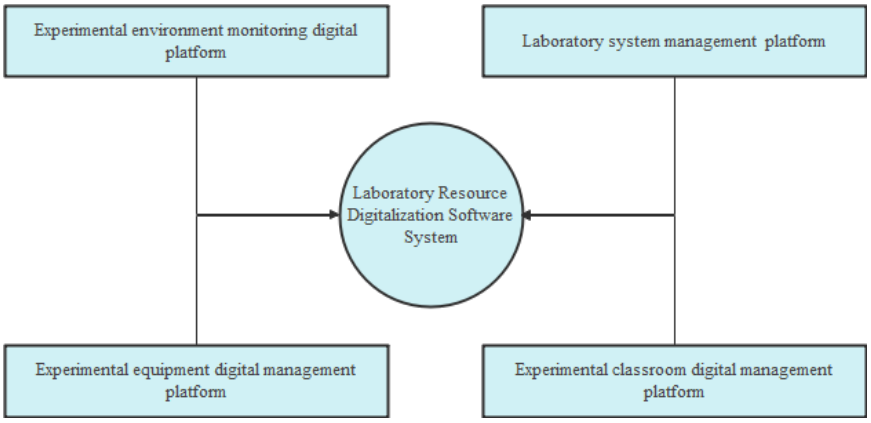

Figure 1: Digitalization Software System structure diagram.

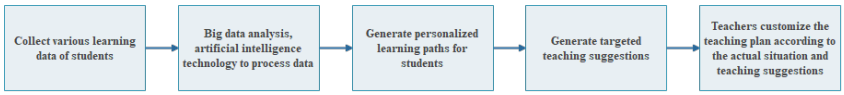

Figure 2: Teaching methods Personalized customized flow charts.

Table 1: Experimental Entity Data.

Entity Name	Entity Attribute
Laboratory Equipment	Equipment Name, Equipment Number, Price, Activation Time
User Personnel	Person's Name, Personnel Category, Personnel Number
Laboratory Classroom	Classroom Name, Classroom Number, Area, Administrator Number
System Administrator	Administrator Account, Name, Password

The middleware and business cluster are used to provide corresponding interface services. The middleware includes components such as RabbitMQ, Spring Cloud Bus, Spring Cloud Task, etc., which can provide microservices interfaces by connecting to the databases[7]. The business cluster includes services for digital equipment, digital classrooms, environmental monitoring data, and system data. The business cluster ensures the scalability and reliability of the modules by encapsulating the functional modules that implement specific business functionalities, while ensuring the provision of all functions required by the upper-layer applications.

The application layer mainly includes web applications[8], mobile terminal applications (Android and iOS), WeChat public accounts, WeChat mini-programs, and other interfaces provided to the outside. On the PC side, users can perform various operations by logging into the digital management platform for laboratory equipment. On the mobile side, users can manage equipment through the laboratory equipment digital management app. At the same time, we also support all operations on the digital management platform for laboratory equipment on WeChat mini-programs or public accounts.

3.2 Implementation of the Digital Teaching Method System

3.2.1 System Framework. The digital system for teaching methods consists of a digital collection module, a digital processing module, and a personalized recommendation generation module, as shown in Figure 4. The digital collection module gathers students' learning data through various channels to provide data source support. The digital processing module utilizes artificial intelligence algorithms to clean the data and extract information related to learning behaviors. Finally, the personalized recommendation generation module creates personalized learning paths based on the processed data through feature matching, meeting the needs of students for personalized learning. It also generates personalized teaching suggestions to help teachers devise teaching plans that are more aligned with the actual conditions of the students.

3.2.2 Algorithm Module. The Isolation Forest algorithm offers an efficient approach to anomaly detection that operates on complete datasets[9]. It boasts a linear time complexity coupled with a high level of precision, making it an ideal solution for identifying anomalies within large volumes of data. This algorithm has found its application across various domains, including the filtration of noisy data, the detection of fraudulent activities in financial transactions, and the identification of diseases[10].The digital processing module

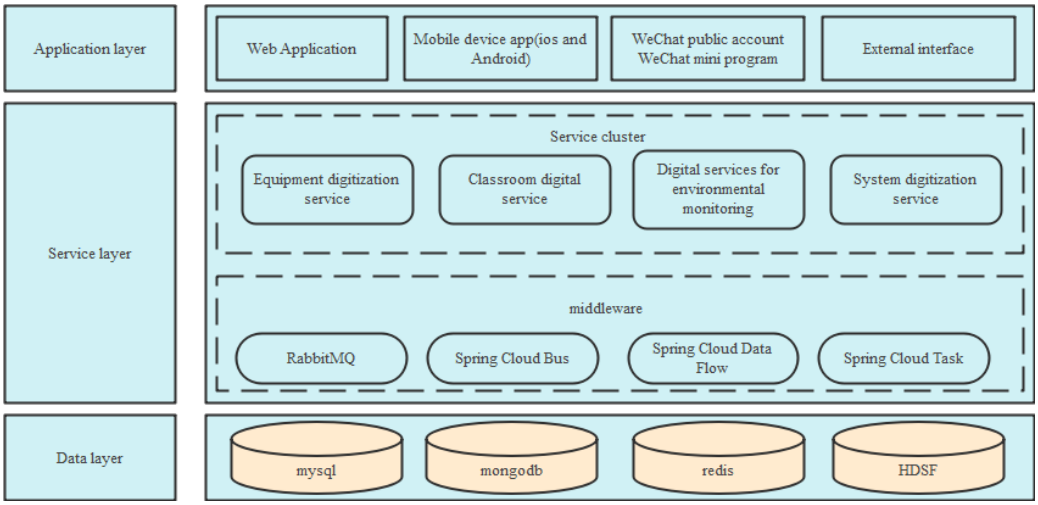

Figure 3: Framework Diagram of the Laboratory Resource Digitalization Software System.

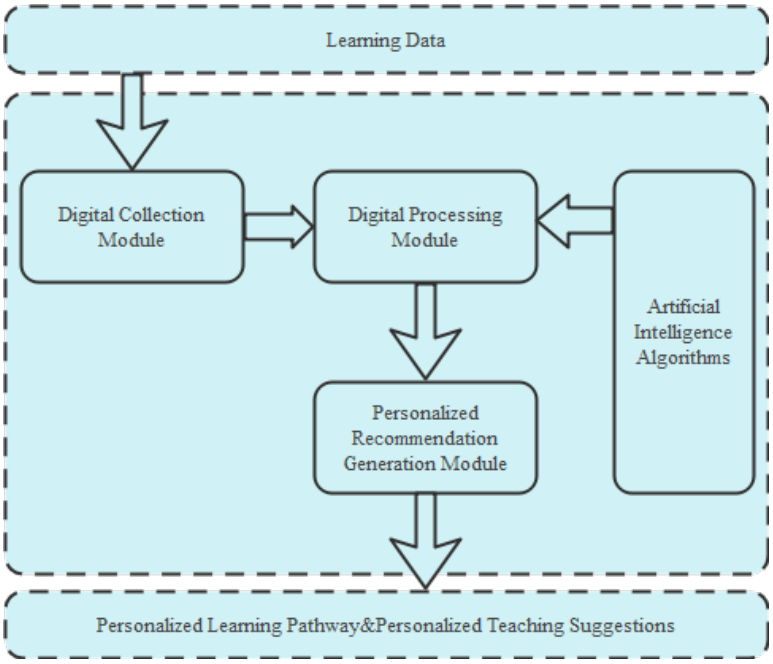

Figure 4: Framework Diagram of the Digital Teaching Methods System.

plays a crucial role in the digital system of teaching methods. The digital processing module first uses the Isolation Forest algorithm to clean the learning data, and then processes the data according to Algorithm 1

The process of Algorithm 1 includes the following steps:

(1)Count the values of various indicators in the indicator library.

(2) Categorize all indicator values according to features.

(3)Label the experimental data based on the classification results.

The pseudo code for Algorithm 1 is presented as follows.

4 Conclusion

In summary, the digital construction of media technology laboratories is an inevitable trend in the development of the digital age, marking that media laboratories have stepped into a brand new digital era in laboratory management and teaching research. The method proposed in this paper for constructing a digital system for laboratory resources and a digital system for teaching methods is conducive to fostering top-notch dynamic innovation capabilities, effectively enhancing the collaborative innovation capabilities of organizations in high-level cutting-edge fields, and improving the

Algorithm 1 Processes data Algorithm

```
Set initial data based on Learning Data while (content is not
finished)
  {
     for each in indicator library
     {
          Get  the sum of indicator values.
for each  in classification set
{
   if indicator matching classification
              {
                   Then let the data be labeled based on the sum of
the indicator values.
              }
}
          }
    Finally output data with labeled features.
    }
```

overall core competitiveness of research teams. This method can effectively integrate originally isolated platforms into the system and through reasonable steps, make the modules more closely integrated, and play a better overall function. Finally, it needs to be pointed out that media technology laboratories need to increase investment in software and hardware in aspects such as digital systems for resources and teaching methods, ultimately achieving comprehensive digitalization, thereby providing strong support for innovative teaching and research in colleges and universities.

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